



**NATIONAL TRANSPORTATION SAFETY BOARD
WASHINGTON, D.C. 20594**

Office of Research and Engineering
Safety Studies and Statistical Analysis Division

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Data Report

Anchor Displacement Data Analysis

A. ACCIDENT INFORMATION

NTSB Accident Number: HWY06MH024
Date of Accident: July 10, 2006
Location of Accident: Boston, Massachusetts
Investigator: Mark Bagnard

B. SAFETY STUDIES AND STATISTICAL ANALYSIS DIVISION

Bruce G. Coury, Ph.D.
Transportation Research Analyst
Safety Studies and Statistical Analysis Division, RE-10
Office of Research and Engineering
Washington, DC 20594

C. PURPOSE OF THE DATA REPORT

This data report analyzes the pattern of displaced anchors presented in the Materials Laboratory Factual Report No. 06-079 for the highway accident investigation HWY06MH024 Connector Tunnel to Ted Williams Tunnel in Boston, Massachusetts. The materials lab report documented the location and amount of downward displacement of anchors that secured concrete panels to the ceiling of highway tunnels. The proportion of displaced anchors was greatest for the highest dead load condition and least for the lowest dead load condition. In addition, there was a lower proportion of displaced anchors in the HOV tunnel than in either the Eastbound or Westbound tunnels.

Specifically, this analysis quantifies the significance of those findings, asking if:

- the frequency of anchor displacement was significantly affected by load; and
- the frequency of anchor displacements differed among the three tunnels.

D. METHODOLOGY

The analysis was conducted in the following way:

1. Determine the pattern of displaced anchors for loading in each of the three tunnels;
2. Select the appropriate statistical analysis; and
3. Determine the expected pattern of displaced anchors based on an assumption that there was no effect of loading or location in a tunnel.

There were a total of 648 anchors in the three tunnels, and within each tunnel, the ceiling system was divided into three rows (North, Middle, and South) corresponding to the estimated loading conditions shown in table 1 (the Eastbound tunnel contained two Middle rows). As a result, anchor location in a tunnel row determined the loading on an anchor. The anchor system was subjected to three types of load: dead, suction, and live. Dead loads were a function of the weight of each structural member of the ceiling system. Suction and live loads were a result of the operation of and maintenance to the tunnel ventilation system. Suction loads were present when the ventilation fans were operating, and live loads were generated when, for example, workers or equipment were present in the ventilation system. Suction loads reduced the total load carried by the anchoring system by essentially creating an uplift load. Live loads, when they occurred, increased the downward load on the anchoring system. For the purposes of this analysis, only the constant dead loads were considered, and the estimated values of those loads divided into four loading conditions are shown in the first column of table 1. Note that the failure of the ceiling system occurred in the Middle 1 row of the Eastbound tunnel.

Table 1: Loading Condition for Each Tunnel Row

	Tunnel Row		
Dead Load (lbs)	HOV	Eastbound	Westbound
27,452		Middle 1	Middle
21,891 to 22,834	Middle	Middle 2	
15,440 to 15,482	North	North	North, South
9,617 to 9,699	South	South	

Of the 648 anchors in the ceiling system, 181 of them (28 percent) exhibited measurable displacement. The number of displaced anchors for each of the four dead load conditions in the three tunnels is shown in table 2 along with the total number of anchors for each tunnel and load condition. Approximately 20 percent of the anchors were subjected to the highest load condition (27,452 pounds) but these anchors accounted for more 50 percent of the displaced anchors.

Table 2: Number of Displaced Anchors by Load in Each Tunnel

	Tunnel			
Dead Load (lbs)	HOV	Eastbound	Westbound	Displaced of Total
27,452	N/A	48 of 70	54 of 70	102 of 140
21,891 to 22,834	11 of 68	14 of 70	N/A	25 of 138
15,440 to 15,482	9 of 60	11 of 64	24 of 122	44 of 246
9,617 to 9,699	6 of 60	4 of 64	N/A	10 of 124
Displaced of Total	26 of 188	77 of 268	78 of 192	181 of 648

D.1 Choice of Statistical Method

Selection of a statistical technique is dependent upon the characteristics of the data and the relations among the variables of interest. On scene work by staff metallurgist resulted in a measure of displacement (in inches) for each anchor. The distribution of that set of data was examined to determine the appropriate analysis. A scatter plot of the measured displacement of the 181 displaced anchors is shown in figure 1. The magnitude of the measured displacement is shown on the y-axis, and the rank order of the magnitude of the displacement is shown on the x-axis. What was evident from this plot was the rapid increase in displacement once 0.5 inch of

displacement was reached. The 20 anchors with a maximum displacement of 5 inches were the ones that failed and represent anchors that were completely displaced from the ceiling system. From a statistical perspective, this graph indicates that the measured displacements were not normally distributed.

In addition, there might have been a dependency between anchors in a row as load was transferred from a displaced anchor to neighboring anchors in that row. Furthermore, the magnitude of displacement could increase over time. Unfortunately, there was no data to relate the magnitude of anchor displacement to time. Such data about the time course of anchor displacement would be necessary for any form of reliability analysis.

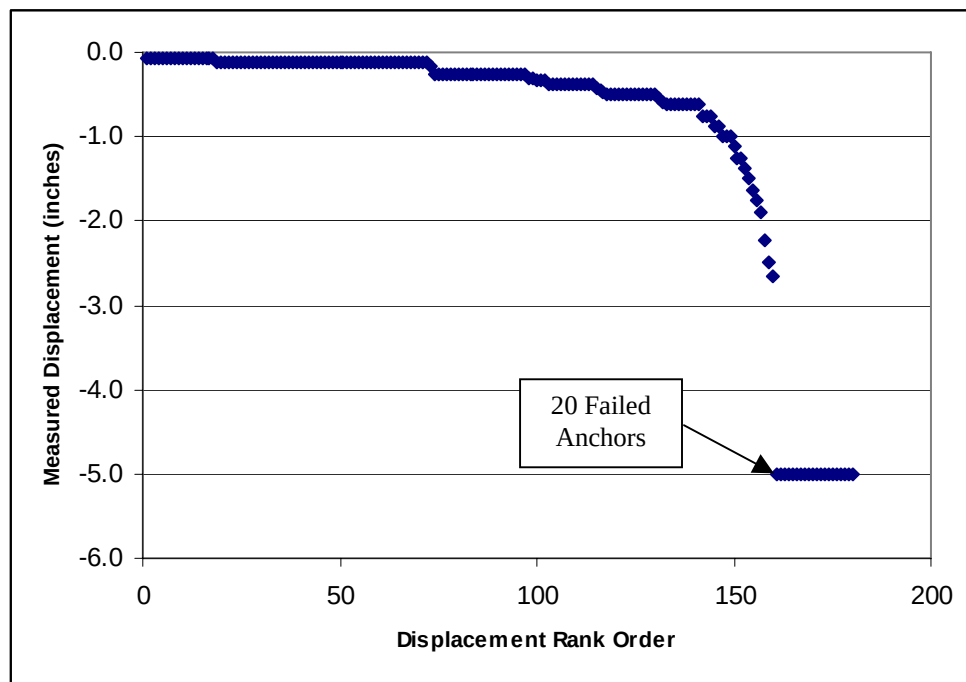


Figure 1: Scatter Plot of Anchor Displacement Values

The lack of normality in the distribution of measured anchor displacements and the potential for time-related anchor displacement effects indicated that a parametric statistical analysis was not appropriate, and that a nonparametric statistical analysis should be used. The Chi-square statistic was chosen for the analysis of the patterns of anchor displacement.

D.2 Analysis

A Chi-square analysis is a nonparametric test of the frequency differences across the levels of a treatment. The chi-square statistic is the sum of the contributions from each of the individual cells in a frequency table. If a given cell differs markedly from an expected frequency

for that cell, then the contribution of that cell to the overall chi-square is large. If a cell is close to the expected frequency for that cell, then the contribution of that cell to the overall chi-square is low. A large chi-square statistic indicates that the observed frequencies differ from the expected frequencies. In the chi-square analysis conducted in this report, there were two treatments: Dead Load and Tunnel. Dead Load had four levels (27,452 pounds, 21,891 – 22,834 pounds, 15,440 – 15,482 pounds, and 9,617 – 9,699 pounds) and Tunnel had three levels (HOV, Eastbound, and Westbound).

There are basically two kinds of chi-square tests. The first is called a one-way analysis, and the second is called a multi-way analysis. The purpose of both is to determine whether the observed frequencies differ from the frequencies that would be expected by chance. However, the multi-way analysis requires that none of the cells in the frequency table are less than five. As shown in table 2, four cells have frequency counts less than 5. Three of the cells are zero (HOV 27,452 pounds, and Westbound 21,891 – 22,834 pounds and 9,617 – 9,699 pounds loading conditions) because there were no anchors in those tunnels carrying that load. As a result, two one-way chi-square analyses were performed: one to analyze the main effect of dead load and another to analyze the main effect of tunnel location. Given that the anchors in the four load conditions and the anchors in the three separate tunnels could be assumed to be independent, the requirements for the one-way chi-square analysis were met. The test of significance was set at a probability less than 0.05.

The analysis began with the assumption that if the effect of dead load or the differences between tunnels was strictly due to chance, then the proportion of displaced anchors would be uniform across the four categories of loads and across the three tunnels. The number of expected displaced anchors for each load in each tunnel is shown in table 3. The values in the table were calculated by determining the proportion of displaced anchors that would have occurred in each of the load and tunnel conditions, given that there was no effect of load or no difference between tunnels. This was done by multiplying the number of total anchors in each cell from table 2 by the total proportion of displaced anchors (27.9 percent). The row and column totals provided the expected frequencies for each chi-square analysis of load and tunnel effects.

Table 3: Expected Number of Displaced Anchors by Load in Each Tunnel

	Tunnel			
Dead Load (lbs)	HOV	Eastbound	Westbound	Expected Load
27,452	0	20	20	40
21,891 to 22,834	19	20	0	39
15,440 to 15,482	17	18	34	69
9,617 to 9,699	17	18	0	35
Expected Tunnel	53	76	54	

E. Results

The analysis showed that both the effect of Dead Load and Tunnel were significant. The results of the chi-square analysis for load is shown in table 4 and tunnel in table 6. In both cases the probability for the test of significance was less than 0.001 (well below the 0.05 criteria), indicating that the deviation of the observed displaced anchor frequencies were significantly different from the expected frequencies.

The analysis of the load effect showed that the highest dead load condition produced more displaced anchors than expected by chance alone, and all of the other loading conditions produced fewer displaced anchors than expected. An additional one-way chi-square analysis that removed the 27,452 pounds loading condition from consideration was conducted. An adjustment to the expected anchor displacement frequencies was made after the 102 displaced anchors in the highest load condition were removed. The adjusted expected frequencies and the results of the analysis are shown in table 5. The analysis found that the differences between the observed and expected frequency of anchor displacement for the three loading conditions approached significance (probability > 0.05), primarily due to the lowest load condition (9,617 – 9,699 pounds) producing fewer displaced anchors than expected. The higher load conditions produced slightly more displaced anchors than expected.

Table 4: One-way Chi-Square Analysis of Load

Dead Load (lbs)	Observed Frequencies	Expected Frequencies	Residual	Test Statistics	
27,452	102	40	62	Chi-square	129.43
21,891 to 22,834	25	39	-14	df	3
15,440 to 15,482	44	69	-25	Significance	<0.001
9,617 to 9,699	10	35	-25		
Total	181				

Table 5: One-way Chi-Square Analysis Excluding Highest Load Condition

Dead Load (lbs)	Observed Frequencies	Expected Frequencies	Residual	Test Statistics	
21,891 to 22,834	25	21	4	Chi-square	5.84
15,440 to 15,482	44	38	6	df	2
9,617 to 9,699	10	19	-9	Significance	0.053
Total	79				

The analysis of the frequency differences between the three tunnels found that the HOV tunnel produced only half of the expected number of displaced anchors, and the Westbound tunnel produced more displaced anchors than expected (table 6). The higher than expected number of displaced anchors in the Westbound tunnel might appear to be an unusual result given that the actual ceiling system failure occurred in the Eastbound tunnel. However, the analysis was based on frequency of displaced anchors, not the magnitude of anchor displacement. When the anchor displacement values shown in figure 1 were related to the Eastbound and Westbound tunnels, 60 percent of anchors with a displacement value of at least 0.5 inch were in the Eastbound tunnel.

Table 6: One-way Chi-Square Analysis of Tunnel

Tunnel	Observed Frequencies	Expected Frequencies	Residual	Test Statistics	
HOV	25	53	-28	Chi-square	24.604
Eastbound	77	75	2	df	2
Westbound	78	54	24	Significance	< 0.001
Total	181				

F. Summary

The analyses conducted in this report confirmed that the highest dead load condition produced significantly more displaced anchors than would be expected from chance alone, and that the differences between the three tunnels were significant. When the highest dead load condition was excluded from the analysis, the number of displaced anchors in the other three loading conditions were not significantly different from the expected frequency. This result suggests that the pattern of displaced anchor frequencies was primarily due to the high dead load condition. The analysis of the tunnels showed that the HOV tunnel produced fewer displaced anchors than expected.